Name: ______________________________

New York State Testing Program

Mathematics Test Session 1

Grade 7

Spring 2026

RELEASED QUESTIONS

Grade 7 Mathematics Reference Sheet

CONVERSIONS

1 yard = 3 feet
1 mile = 5,280 feet

1 cup = 8 fluid ounces
1 pint = 2 cups
1 quart = 2 pints
1 gallon = 4 quarts

1 pound = 16 ounces
1 ton = 2,000 pounds

CONVERSIONS ACROSS MEASUREMENT SYSTEMS

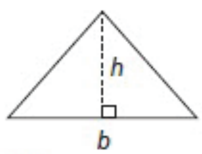
1 inch = 2.54 centimeters
1 meter = 39.37 inches
1 mile = 1.609 kilometers
1 kilometer = 0.6214 mile

1 gallon = 3.785 liters
1 liter = 0.2642 gallon

1 pound = 0.454 kilogram
1 kilogram = 2.2 pounds

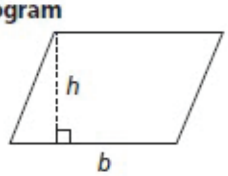
FORMULAS AND FIGURES

Triangle



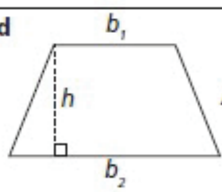
$A = \frac{1}{2}bh$

Parallelogram



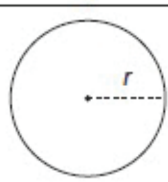
$A = bh$

Trapezoid



$A = \frac{1}{2}h(b_1 + b_2)$

Circle



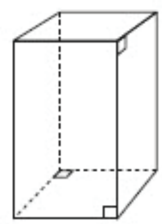
$C = 2\pi r$
$C = \pi d$
$A = \pi r^2$

Simple Interest

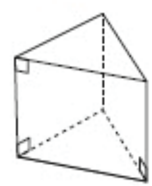
$I = prt$ where I is interest,
p is principal,
r is rate, and
t is time

General Prism $V = Bh$

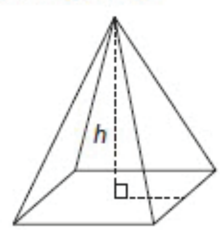
Right Rectangular Prism



Right Triangular Prism



Right Rectangular Pyramid



Session 1

TIPS FOR TAKING THE TEST

Here are some ideas to help you do your best:

- Read each question carefully. Take your time.
- You have a ruler, a protractor, a reference sheet, and a calculator that you can use on the test if they help you answer the question.

1 Mr. Joseph paid a total of $\$540.00$ for a new gaming system and games. He paid $\$300.00$ for the gaming system and $\$60.00$ for each game. How many games did Mr. Joseph buy?

A 3

B 4

C 5

D 9

2 A group of four students is playing a game using a deck of cards. The cards in the deck are labeled with an integer from -10 to 10. Each student draws two cards from the deck and finds the sum of their two cards to determine their score. The two cards drawn by each student are shown below.

Alberto		**Jenny**	
–4	3	–5	5

Lisako		**Paul**	
–2	–2	1	–1

Which students have a score of 0 ?

A Alberto and Jenny

B Jenny and Lisako

C Alberto and Paul

D Jenny and Paul

GO ON

3 A farm sells blueberries by the pound to customers. The table below shows the relationship between the price, p, in dollars, and the number of pounds of blueberries, b, sold.

BLUEBERRY PRICES

Pounds of Blueberries, *b*	Price (dollars), *p*
2	$11.98
4	$23.96
6	$35.94

Which statement describes a constant of proportionality for the relationship between the price, p, in dollars, and the number of pounds of blueberries, b, sold?

A The price per pound is $\$0.17$.

B The price per pound is $\$5.99$.

C The price per pound is $\$9.98$.

D The price per pound is $\$11.98$.

4 What is the value of the expression $\left(-\frac{5}{6}\right) \div \left(-\frac{1}{3}\right)$?

A $-2\frac{1}{2}$

B $-\frac{5}{18}$

C $\frac{5}{18}$

D $2\frac{1}{2}$

GO ON

5 A water hose is being used to fill a $3{,}000$ gallon swimming pool. Water flows at a constant rate through the hose. If the pool is completely filled in $2\frac{1}{2}$ hours, how many gallons of water were added to the pool per hour?

A 300

B 600

C $1{,}200$

D $7{,}500$

6 Shameka is saving for a new scooter that costs $\$248.00$. She has already saved $\$100.00$. She wants to buy the scooter in 8 weeks by saving the same amount of money each week. What is the least amount of money, in dollars, Shameka can save each week to achieve her goal?

A $\$16.45$

B $\$18.50$

C $\$31.00$

D $\$43.50$

GO ON

7 The table below shows a proportional relationship between y and x.

x	y
2	0.6
3	0.9
4	1.2
5	1.5

Which equation represents the proportional relationship?

A $y = x + 0.3$

B $y = x + 3.\overline{3}$

C $y = 3.\overline{3}x$

D $y = 0.3x$

8 A student can choose one food item and one type of juice for breakfast. The food items and types of juice available are listed below.

- food item: egg, toast, pancakes, sausage
- type of juice: apple, grape, orange

How many different breakfast combinations of one food item and one type of juice are available for the student to choose from?

A 3

B 4

C 7

D 12

GO ON

10 Ms. Morgan earns $15.50 per hour working in an office. On Tuesday she spends $1\frac{1}{3}$ hours filling out paperwork, 1 hour and 25 minutes sending emails, and $3\frac{1}{4}$ hours making appointments. How much money does Ms. Morgan earn on Tuesday?

A $77.50

B $89.90

C $93.00

D $108.50

GO ON

11 Which expression is equivalent to $3(5x+2)-2(6x-7)$?

A $23x$

B $-11x$

C $3x-5$

D $3x+20$

12 Jessica opened a savings account with a $\$1{,}200.00$ deposit. She did not make any additional deposits or withdrawals for 4 years and earned $\$144.00$ in simple interest. What was the annual interest rate that the bank used to calculate the simple interest?

A 0.3%

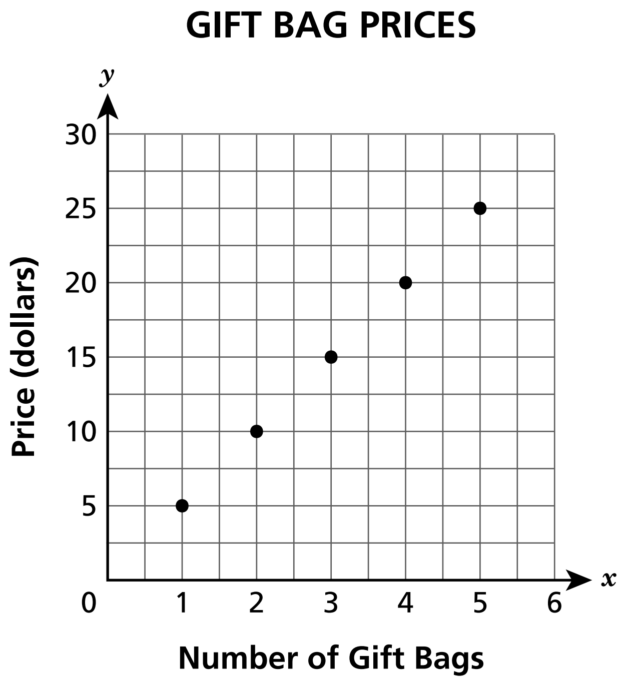
B 2.7%

C 3.0%

D 4.8%

GO ON

13 The graph below shows the prices of gift bags at a store.



Based on the graph, which statement is true?

A The unit rate is $0.20 per gift bag.

B The unit rate is $1.00 per gift bag.

C The unit rate is $2.00 per gift bag.

D The unit rate is $5.00 per gift bag.

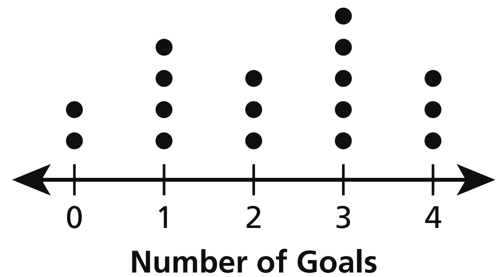
GO ON

14 The number of goals scored by two soccer teams in each of 17 games during a tournament is shown in the table and dot plot below.

TEAM A

Number of Goals	Frequency
0	2
1	4
2	2
3	3
4	6

TEAM B



Which team had a higher median number of goals scored in a game and what is that team's median number of goals scored?

A Team A with a median of 3 goals scored in a game.

B Team A with a median of 4 goals scored in a game.

C Team B with a median of 2 goals scored in a game.

D Team B with a median of 3 goals scored in a game.

15 What is the product of $-\frac{1}{5}$ and $\frac{3}{8}$?

A $-\frac{8}{15}$

B $-\frac{3}{40}$

C $\frac{3}{40}$

D $\frac{8}{15}$

GO ON

17 Joyce is buying a laptop. The original price of the laptop is $279.00. She uses a coupon for a 10% discount off the original price. A sales tax of 8.875% is applied to the discounted price. How much does Joyce pay, in total, for the laptop, rounded to the nearest cent?

A $228.81

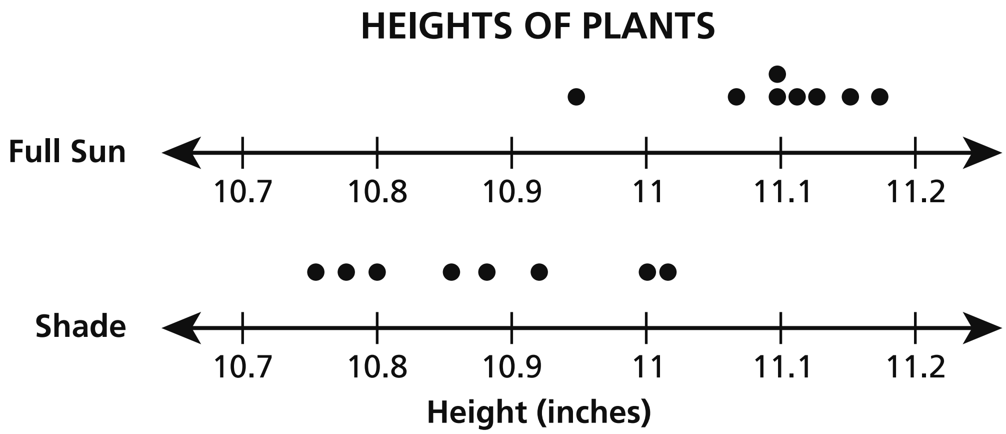
B $260.13

C $273.39

D $334.14

GO ON

18 In a science class, students placed 8 tomato plants in full sun and 8 tomato plants in the shade. The heights of the plants were measured and recorded after several weeks. The results are shown in the dot plots below.



Which statement **best** describes the data?

A The heights of the plants in full sun varied more than the heights of the plants in the shade.

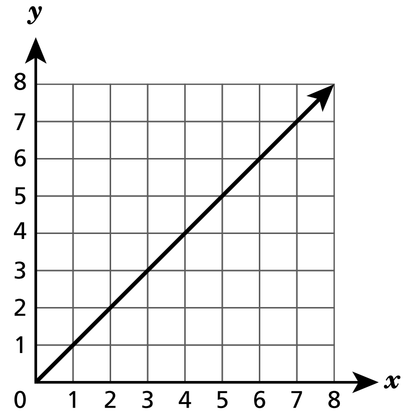
B The heights of the plants in the shade were all greater than the heights of the plants in full sun.

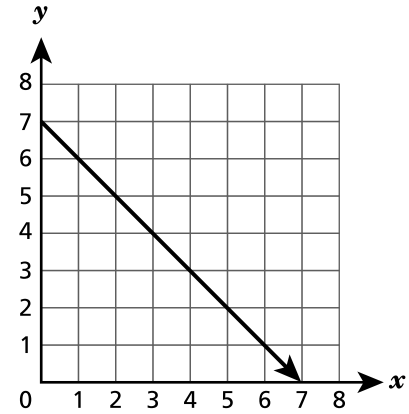
C The average height of the plants in full sun was greater than the average height of the plants in the shade.

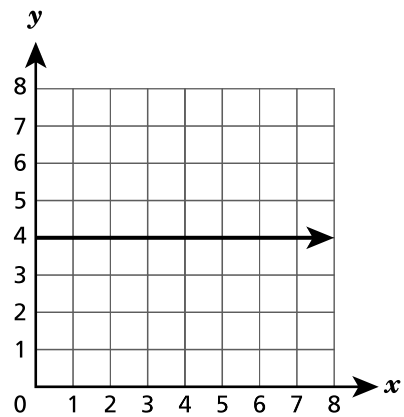
D The average height of the plants in the shade was greater than the average height of the plants in full sun.

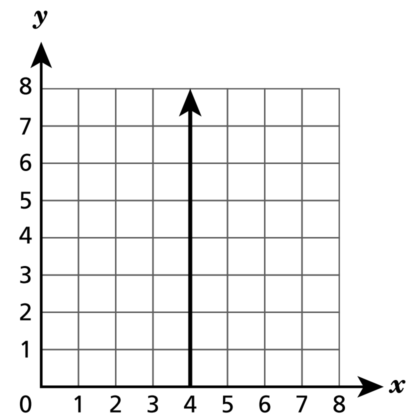
GO ON

19 Which graph represents a proportional relationship?

A 

C 

B 

D 

20 An equation is shown below.

$$4(x + 2) = \frac{12}{-0.25}$$

What value of x makes the equation true?

A -12.5

B -14

C -50

D -56

GO ON

21 An artist is paid $\$20.00$ to work at a birthday party, plus $\$5.00$ for each portrait painted during the party. The artist would like to earn at least $\$200.00$ at the party. Which inequality represents the number of portraits, x, the artist should paint to earn at least $\$200.00$?

A $20x + 5 \leq 200$

B $20x + 5 < 200$

C $5x + 20 \geq 200$

D $5x + 20 > 200$

22 Bianca created a scale drawing of her bedroom for a class project using a scale of 1 ft to 1.5 cm. On the drawing, Bianca's rectangular bed is 10.5 cm long and 5.25 cm wide. What is the actual area, in square feet, of Bianca's bed?

A 24.5

B 36.75

C 55.125

D 124.03

23 Which expression is equivalent to $7 - 2(3x + 6) - 4x$?

A $11x + 6$

B $11x + 30$

C $-10x - 5$

D $-10x + 13$

GO ON

25 Mateo has $\$20.00$ to spend at a store. He spends $\frac{2}{5}$ of his money on a shirt. Then he buys a pack of gum with 10% of his remaining money. What is the total amount of money that Mateo spends at the store?

A $\$8.80$

B $\$9.20$

C $\$10.80$

D $\$13.20$

GO ON

26 What is the value of $\left(\frac{4}{5}\right)(0.2)\left(-\frac{5}{8}\right)$?

A $-\frac{1}{10}$

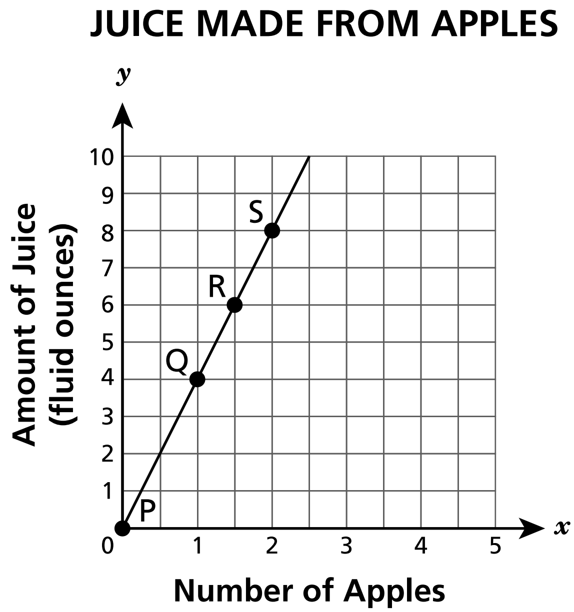
B $-\frac{3}{8}$

C $\frac{1}{10}$

D $\frac{3}{8}$

GO ON

27 The graph below shows the relationship between the amount of juice, y, that can be made from x apples.



Which point on the graph has a y coordinate that represents the number of fluid ounces of juice made from 1 apple?

A point P

B point Q

C point R

D point S

GO ON

29 Students in four middle school classes raised money for a field trip. The amount of money each class raised is described below.

- Ms. Strazullo's class raised $153.45.
- Mr. Fiorentino's class raised $123.23 more than Ms. Strazullo's class.
- Ms. Rubas' class raised $176.32 less than Mr. Fiorentino's class.
- Mr. Smith's class raised $234.89 more than Ms. Rubas' class.

If the total price of the field trip is $800.00, how much extra money did they raise?

A $65.74

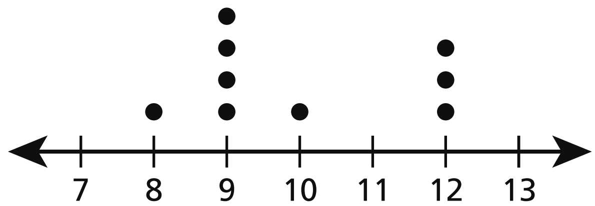
B $112.11

C $687.89

D $865.74

GO ON

30 The weights of 9 adult cats are recorded on the dot plot shown below.



Which label for the number line could be used to correctly identify the unit of measurement for the data in the dot plot?

A number of cats

B number of years

C number of inches

D number of pounds

31 An incomplete equation is shown below.

$$(5.2 - 1.6) + (\underline{\quad ? \quad}) = 0$$

Which expression completes the equation to make it true?

A $5.2 + 1.6$

B $5.2 - 1.6$

C $-5.2 - 1.6$

D $-5.2 + 1.6$

GO ON

32 Which expression is equivalent to $-\frac{1}{4}(12x - 6)$?

A $3x + 6$

B $3x - 6$

C $-3x + 1\frac{1}{2}$

D $-3x - 1\frac{1}{2}$

STOP

Session 2

TIPS FOR TAKING THE TEST

Here are some ideas to help you do your best:

- Read each question carefully. Take your time.
- You have a ruler, a protractor, a reference sheet, and a calculator that you can use on the test if they help you answer the question.
- Be sure to show your work when asked.
- Be sure to explain your answer when asked.

33 The table below shows the number of gallons of gasoline customers put in their cars and the total price they paid.

GASOLINE PRICES

Gallons, *g*	Total Price, *p*
3	$9.63
5	$16.05
9	$28.89

Which equation represents the relationship between the total price, p, and gallons of gasoline purchased, g?

A $p = 0.31g$

B $p = 3.21g$

C $p = 6.42g$

D $p = 9.63g$

34 Which expression is equivalent to $\left(\frac{-45}{-9}\right)$?

A $-\left(\frac{45}{9}\right)$

B $\frac{-45}{9}$

C $\frac{45}{-9}$

D $\frac{45}{9}$

GO ON

35 Ms. Suarez buys 4 packages of pencils for $\$4.48$. Each package of pencils is the same price. What is the unit price, in dollars per package, of the pencils?

A $\$0.48$

B $\$0.89$

C $\$1.12$

D $\$4.00$

36 Ashley buys a book that is on sale for 25% off the original price, x. The expression $x - 0.25x$ can be used to determine the final price of the book. Which expression can also be used to determine the final price of the book?

A $0.25x$

B $0.75x$

C $0.25x - x$

D $0.75x + x$

GO ON

37 Jayden made a scale drawing of the rectangular basketball court at his middle school.

- The drawing has a scale of 1 inch to 24 feet.
- The actual length of the basketball court is $76\frac{4}{5}$ feet.

What is the length, in inches, of the rectangular basketball court in Jayden's drawing?

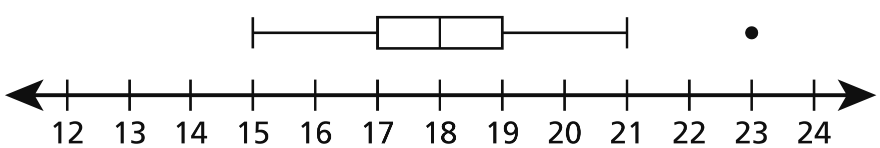
A $\frac{5}{16}$

B $3\frac{1}{5}$

C 24

D $52\frac{1}{5}$

38 A box plot is shown below.



Based on the box plot, what is true about the data value 23 ?

A It represents a data point that lies an unusual distance from other points in the data.

B It represents the data point that occurred the most number of times in the data.

C It represents the interquartile range of the data.

D It represents the median of the data.

GO ON

40 **This question is worth 1 credit.**

Mr. Morgano goes to the grocery store. The prices of different items are shown in the table below.

GROCERY ITEM PRICES

Item	Price
One pound of grapes	\$1.59
One pound of turkey	\$8.24
One loaf of bread	\$1.33
One gallon of chocolate milk	\$2.78

Mr. Morgano buys 3 pounds of grapes, $\frac{1}{2}$ pound of turkey, 2 loaves of bread, and 1 gallon of chocolate milk. He pays with a $\$20.00$ bill. How much change, in dollars, should Mr. Morgano receive?

Answer $ ________________

GO ON

41 **This question is worth 1 credit.**

Simplify the expression shown below completely.

$$-\frac{3}{4}(4x - 8) + (x - 2)$$

Answer ________________

GO ON

43 **This question is worth 2 credits.**

A customer orders a large pizza with 2 toppings and a medium pizza with 2 toppings at a restaurant. The prices for the pizzas and toppings are listed below.

- Large pizza without toppings: $12.50
- Medium pizza without toppings: $10.25
- Toppings: $1.25 each

A 20% discount is applied to the customer's order. What is the total price of the order, excluding tax, after the discount?

Show your work.

Answer $ ________________

GO ON

44 **This question is worth 2 credits.**

Jim and Devin ride their bicycles to meet at a friend's house.

- Jim rides $3\frac{1}{3}$ miles in $\frac{1}{2}$ hour.
- Devin rides $1\frac{7}{8}$ miles in $\frac{1}{4}$ hour.

What is the difference, in miles per hour, between the average speeds of Jim and Devin?

Explain how you determined your answer.

___

___

___

GO ON

45 **This question is worth 2 credits.**

A group of sixth grade students recorded their favorite flavors of ice cream. The results are shown in the table below.

FAVORITE FLAVORS OF ICE CREAM

Flavor of Ice Cream	Number of Students
Vanilla	12
Chocolate	8
Strawberry	5
Pistachio	3
Other	4

Based on the results, what would be the predicted number of students in the entire sixth grade whose favorite ice cream flavor is chocolate, if there are a total of 120 students in the sixth grade?

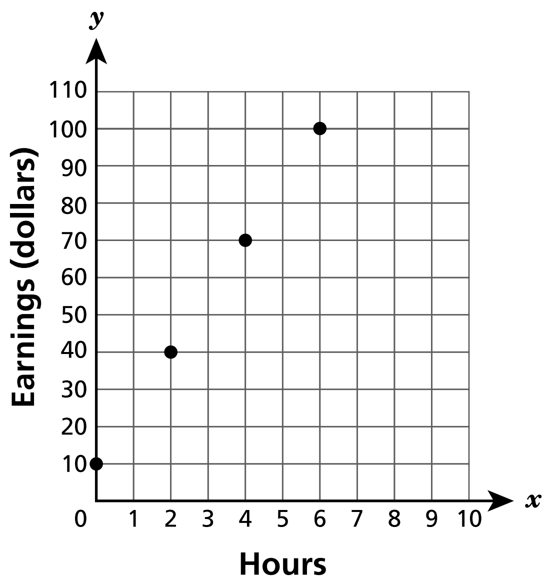
Show your work.

Answer ________________ students

46 **This question is worth 2 credits.**

Eric's and Jenna's earnings at their jobs are shown in the graph and the table below.

ERIC'S EARNINGS



JENNA'S EARNINGS

Hours, x	Earnings (dollars), y
3	54
6	108
9	162
12	216

For which person is the amount of earnings proportional to the number of hours worked?

Explain your answer.

____

____

____

GO ON

47 **This question is worth 2 credits.**

An expression is shown below.

$$\left(\frac{1}{2}\right)(-0.4) \div \left(\frac{1}{3}\right)$$

Evaluate the expression.

Show your work.

Answer ________________

GO ON

48 **This question is worth 3 credits.**

Nicholas went to a bowling alley. The bowling alley charged Nicholas a one time fee of $\$3.75$ to rent shoes and $\$5.25$ for each game he played. Write and solve an equation to determine the number of games, x, that Nicholas played if he spent a total of $\$30.00$ on shoes and games.

Show your work.

Answer ________________ games

STOP

THE STATE EDUCATION DEPARTMENT

THE UNIVERSITY OF THE STATE OF NEW YORK / ALBANY, NY 12234

2026 Mathematics Tests Map to the Standards

Grade 7

Question	Type	Key	Points	Standard	Cluster	Subscore	Secondary Standard(s)
Session 1							
1	Multiple Choice	B	1	NGLS.Math.Content.NY-7.EE.4a	Expressions and Equations	Expressions and Equations	
2	Multiple Choice	D	1	NGLS.Math.Content.NY-7.NS.1b	The Number System	The Number System	
3	Multiple Choice	B	1	NGLS.Math.Content.NY-7.RP.2b	Ratios and Proportional Relationships	Ratios and Proportional Relationships	
4	Multiple Choice	D	1	NGLS.Math.Content.NY-7.NS.2c	The Number System	The Number System	
5	Multiple Choice	C	1	NGLS.Math.Content.NY-7.RP.1	Ratios and Proportional Relationships	Ratios and Proportional Relationships	
6	Multiple Choice	B	1	NGLS.Math.Content.NY-7.EE.4b	Expressions and Equations	Expressions and Equations	
7	Multiple Choice	D	1	NGLS.Math.Content.NY-7.RP.2c	Ratios and Proportional Relationships	Ratios and Proportional Relationships	
8	Multiple Choice	D	1	NGLS.Math.Content.NY-7.SP.8b	Statistics and Probability		
10	Multiple Choice	C	1	NGLS.Math.Content.NY-7.EE.3	Expressions and Equations	Expressions and Equations	
11	Multiple Choice	D	1	NGLS.Math.Content.NY-7.EE.1	Expressions and Equations	Expressions and Equations	
12	Multiple Choice	C	1	NGLS.Math.Content.NY-7.RP.3	Ratios and Proportional Relationships	Ratios and Proportional Relationships	
13	Multiple Choice	D	1	NGLS.Math.Content.NY-7.RP.2d	Ratios and Proportional Relationships	Ratios and Proportional Relationships	
14	Multiple Choice	A	1	NGLS.Math.Content.NY-7.SP.4	Statistics and Probability		
15	Multiple Choice	B	1	NGLS.Math.Content.NY-7.NS.2c	The Number System	The Number System	
17	Multiple Choice	C	1	NGLS.Math.Content.NY-7.RP.3	Ratios and Proportional Relationships	Ratios and Proportional Relationships	
18	Multiple Choice	C	1	NGLS.Math.Content.NY-7.SP.3	Statistics and Probability		
19	Multiple Choice	A	1	NGLS.Math.Content.NY-7.RP.2a	Ratios and Proportional Relationships	Ratios and Proportional Relationships	
20	Multiple Choice	B	1	NGLS.Math.Content.NY-7.EE.4a	Expressions and Equations	Expressions and Equations	
21	Multiple Choice	C	1	NGLS.Math.Content.NY-7.EE.4b	Expressions and Equations	Expressions and Equations	
22	Multiple Choice	A	1	NGLS.Math.Content.NY-7.G.1	Geometry		
23	Multiple Choice	C	1	NGLS.Math.Content.NY-7.EE.1	Expressions and Equations	Expressions and Equations	
25	Multiple Choice	B	1	NGLS.Math.Content.NY-7.EE.3	Expressions and Equations	Expressions and Equations	
26	Multiple Choice	A	1	NGLS.Math.Content.NY-7.NS.3	The Number System	The Number System	
27	Multiple Choice	B	1	NGLS.Math.Content.NY-7.RP.2d	Ratios and Proportional Relationships	Ratios and Proportional Relationships	
29	Multiple Choice	A	1	NGLS.Math.Content.NY-7.NS.3	The Number System	The Number System	
30	Multiple Choice	D	1	NGLS.Math.Content.NY-6.SP.5b	Statistics and Probability		
31	Multiple Choice	D	1	NGLS.Math.Content.NY-7.NS.1a	The Number System	The Number System	
32	Multiple Choice	C	1	NGLS.Math.Content.NY-7.EE.1	Expressions and Equations	Expressions and Equations	
Session 2							
33	Multiple Choice	B	1	NGLS.Math.Content.NY-7.RP.2c	Ratios and Proportional Relationships	Ratios and Proportional Relationships	
34	Multiple Choice	D	1	NGLS.Math.Content.NY-7.NS.2b	The Number System	The Number System	
35	Multiple Choice	C	1	NGLS.Math.Content.NY-7.RP.2b	Ratios and Proportional Relationships	Ratios and Proportional Relationships	
36	Multiple Choice	B	1	NGLS.Math.Content.NY-7.EE.2	Expressions and Equations	Expressions and Equations	
37	Multiple Choice	B	1	NGLS.Math.Content.NY-7.G.1	Geometry		
38	Multiple Choice	A	1	NGLS.Math.Content.NY-7.SP.1	Statistics and Probability		
40	Constructed Response		1	NGLS.Math.Content.NY-7.NS.3	The Number System	The Number System	
41	Constructed Response		1	NGLS.Math.Content.NY-7.EE.1	Expressions and Equations	Expressions and Equations	
43	Constructed Response		2	NGLS.Math.Content.NY-7.EE.3	Expressions and Equations	Expressions and Equations	NGLS.Math.Content.NY-7.RP.3
44	Constructed Response		2	NGLS.Math.Content.NY-7.RP.1	Ratios and Proportional Relationships	Ratios and Proportional Relationships	
45	Constructed Response		2	NGLS.Math.Content.NY-6.SP.7	Statistics and Probability		
46	Constructed Response		2	NGLS.Math.Content.NY-7.RP.2a	Ratios and Proportional Relationships	Ratios and Proportional Relationships	
47	Constructed Response		2	NGLS.Math.Content.NY-7.NS.2c	The Number System	The Number System	
48	Constructed Response		3	NGLS.Math.Content.NY-7.EE.4a	Expressions and Equations	Expressions and Equations	

*This item map is intended to identify the primary analytic skills necessary to successfully answer each question. However, some questions measure proficiencies described in multiple standards, including a balanced combination of procedural and conceptual understanding.

1-Credit Constructed-Response Rubric

1 Credit	A 1-credit response is a **correct answer** to the question which indicates a thorough understanding of mathematical concepts and/or procedures.
0 Credits*	A 0-credit response is incorrect, irrelevant, or incoherent.

* Condition Code A is applied whenever a student who is present for a test session leaves an entire constructed-response question in that session completely blank (no response attempted).

2-Credit Constructed-Response Holistic Rubric

2 Credits	A 2-credit response includes the correct solution to the question and demonstrates a thorough understanding of the mathematical concepts and/or procedures in the task. This response • indicates that the student has completed the task correctly, using mathematically sound procedures • contains sufficient work to demonstrate a thorough understanding of the mathematical concepts and/or procedures • may contain inconsequential errors that do not detract from the correct solution and the demonstration of a thorough understanding
1 Credit	A 1-credit response demonstrates only a partial understanding of the mathematical concepts and/or procedures in the task. This response • correctly addresses only some elements of the task • may contain an incorrect solution but applies a mathematically appropriate process • may contain the correct solution but required work is incomplete
0 Credits*	A 0-credit response is incorrect, irrelevant, incoherent, or contains a correct solution obtained using an obviously incorrect procedure. Although some elements may contain correct mathematical procedures, holistically they are not sufficient to demonstrate even a limited understanding of the mathematical concepts embodied in the task.

* Condition Code A is applied whenever a student who is present for a test session leaves an entire constructed-response question in that session completely blank (no response attempted).

3-Credit Constructed-Response Holistic Rubric

3 Credits	A 3-credit response includes the correct solution(s) to the question and demonstrates a thorough understanding of the mathematical concepts and/or procedures in the task. This response • indicates that the student has completed the task correctly, using mathematically sound procedures • contains sufficient work to demonstrate a thorough understanding of the mathematical concepts and/or procedures • may contain inconsequential errors that do not detract from the correct solution(s) and the demonstration of a thorough understanding
2 Credits	A 2-credit response demonstrates a partial understanding of the mathematical concepts and/or procedures in the task. This response • appropriately addresses most but not all aspects of the task using mathematically sound procedures • may contain an incorrect solution but provides sound procedures, reasoning, and/or explanations • may reflect some minor misunderstanding of the underlying mathematical concepts and/or procedures
1 Credit	A 1-credit response demonstrates only a limited understanding of the mathematical concepts and/or procedures in the task. This response • may address some elements of the task correctly but reaches an inadequate solution and/or provides reasoning that is faulty or incomplete • exhibits multiple flaws related to misunderstanding of important aspects of the task, misuse of mathematical procedures, or faulty mathematical reasoning • reflects a lack of essential understanding of the underlying mathematical concepts • may contain the correct solution(s) but required work is limited
0 Credits*	A 0-credit response is incorrect, irrelevant, incoherent, or contains a correct solution obtained using an obviously incorrect procedure. Although some elements may contain correct mathematical procedures, holistically they are not sufficient to demonstrate even a limited understanding of the mathematical concepts embodied in the task.

* Condition Code A is applied whenever a student who is present for a test session leaves an entire constructed-response question in that session completely blank (no response attempted).

1-Credit Constructed-Response Mathematics Scoring Policies

1. The student is **not** required to show work for a 1-credit constructed-response question, therefore, any work shown will **not** be scored. A clearly identified correct response should still receive full credit.
2. If the student clearly identifies a correct answer but fails to write that answer in the answer space, the student should still receive full credit.
3. If the student provides one legible response (and one response only), the rater should score the response, even if it has been crossed out.
4. If the student has written more than one response but has crossed some out, the rater should score only the response that has **not** been crossed out.
5. If the student provides more than one response but does not indicate which response is to be considered the correct response and none have been crossed out, the student shall not receive credit.
6. If the student does not provide the answer in the form as directed in the question, the student will not receive credit.
7. In questions requiring number sentences, the number sentences must be written horizontally.
8. When measuring angles with a protractor, there is a +/- 5 degrees deviation allowed of the true measure.
9. Condition Code A is applied whenever a student who is present for a test session leaves an entire constructed-response question in that session completely blank (no response attempted). This is not to be confused with a score of zero wherein the student does respond to part or all of the question, but that work results in a score of zero.

2- and 3-Credit Constructed-Response Mathematics Scoring Policies

1. If a student shows the work in other than a designated "Show your work" or "Explain" area, that work should still be scored.
2. If the question requires students to show their work, and the student shows appropriate work and clearly identifies a correct answer but fails to write that answer in the answer space, the student should still receive full credit.
3. If students are directed to show work or provide an explanation, a correct answer with **no** work shown or **no** explanation provided, receives **no** credit.
4. If students are **not** directed to show work, any work shown will **not** be scored. This applies to questions that do **not** ask for any work and questions that ask for work for one part and do **not** ask for work in another part.
5. If the student provides one legible response (and one response only), the rater should score the response, even if it has been crossed out.
6. If the student has written more than one response but has crossed some out, the rater should score only the response that has **not** been crossed out.
7. If the student provides more than one response, but does not indicate which response is to be considered the correct response and none have been crossed out, the student shall not receive full credit.
8. Trial-and-error responses are **not** subject to Scoring Policy #6 above, since crossing out is part of the trial-and-error process.
9. If a response shows repeated occurrences of the same conceptual error within a question, the conceptual error should **not** be considered more than once in gauging the demonstrated level of understanding.
10. In questions requiring number sentences, the number sentences must be written horizontally.
11. When measuring angles with a protractor, there is a +/- 5 degrees deviation allowed of the true measure.
12. Condition Code A is applied whenever a student who is present for a test session leaves an entire constructed-response question in that session completely blank (no response attempted). This is not to be confused with a score of zero wherein the student does respond to part or all of the question but that work results in a score of zero.

40

Mr. Morgano goes to the grocery store. The prices of different items are shown in the table below.

GROCERY ITEM PRICES

Item	Price
One pound of grapes	\$1.59
One pound of turkey	\$8.24
One loaf of bread	\$1.33
One gallon of chocolate milk	\$2.78

Mr. Morgano buys 3 pounds of grapes, $\frac{1}{2}$ pound of turkey, 2 loaves of bread, and 1 gallon of chocolate milk. He pays with a \$20.00 bill. How much change, in dollars, should Mr. Morgano receive?

Answer \$ ____________

EXEMPLARY RESPONSE

40

Mr. Morgano goes to the grocery store. The prices of different items are shown in the table below.

GROCERY ITEM PRICES

Item	Price
One pound of grapes	\$1.59
One pound of turkey	\$8.24
One loaf of bread	\$1.33
One gallon of chocolate milk	\$2.78

Mr. Morgano buys 3 pounds of grapes, $\frac{1}{2}$ pound of turkey, 2 loaves of bread, and 1 gallon of chocolate milk. He pays with a \$20.00 bill. How much change, in dollars, should Mr. Morgano receive?

Answer \$ 5.67

GUIDE PAPER 1

40

Mr. Morgano goes to the grocery store. The prices of different items are shown in the table below.

GROCERY ITEM PRICES

Item	Price
One pound of grapes	$1.59
One pound of turkey	$8.24
One loaf of bread	$1.33
One gallon of chocolate milk	$2.78

Mr. Morgano buys 3 pounds of grapes, $\frac{1}{2}$ pound of turkey, 2 loaves of bread, and 1 gallon of chocolate milk. He pays with a $20.00 bill. How much change, in dollars, should Mr. Morgano receive?

Answer $ 5.67

Score Credit 1 (out of 1 credit)

A correct answer is provided.

GUIDE PAPER 2

40

Mr. Morgano goes to the grocery store. The prices of different items are shown in the table below.

GROCERY ITEM PRICES

Item	Price
One pound of grapes	$1.59
One pound of turkey	$8.24
One loaf of bread	$1.33
One gallon of chocolate milk	$2.78

Mr. Morgano buys 3 pounds of grapes, $\frac{1}{2}$ pound of turkey, 2 loaves of bread, and 1 gallon of chocolate milk. He pays with a $20.00 bill. How much change, in dollars, should Mr. Morgano receive?

Answer $ 5.67

Score Credit 1 (out of 1 credit)

A correct answer is provided.

GUIDE PAPER 3

40

Mr. Morgano goes to the grocery store. The prices of different items are shown in the table below.

GROCERY ITEM PRICES

Item	Price
One pound of grapes	$1.59
One pound of turkey	$8.24
One loaf of bread	$1.33
One gallon of chocolate milk	$2.78

Mr. Morgano buys 3 pounds of grapes, $\frac{1}{2}$ pound of turkey, 2 loaves of bread, and 1 gallon of chocolate milk. He pays with a $20.00 bill. How much change, in dollars, should Mr. Morgano receive?

Answer $ 5.70

Score Credit 0 (out of 1 credit)

An incorrect answer is provided.

41

Simplify the expression shown below completely.

$$-\frac{3}{4}(4x - 8) + (x - 2)$$

Answer ________________

EXEMPLARY RESPONSE

41

Simplify the expression shown below completely.

$$-\frac{3}{4}(4x - 8) + (x - 2)$$

Answer $-2x + 4$

OR equivalent

41

Simplify the expression shown below completely.

$$-\frac{3}{4}(4x-8)+(x-2)$$

Answer -2x+4

Score Credit 1 (out of 1 credit)

A correct answer is provided.

GUIDE PAPER 2

41

Simplify the expression shown below completely.

$$-\frac{3}{4}(4x - 8) + (x - 2)$$

$$-\frac{3}{4}(4x-8)+(x-2)$$

$$-3x\boxed{+6}+x\boxed{-2}$$

$$-2x+4$$

Answer

Score Credit 1 (out of 1 credit)

A correct answer is provided.

41

Simplify the expression shown below completely.

$$-\frac{3}{4}(4x - 8) + (x - 2)$$

Answer 2x+4

Score Credit 0 (out of 1 credit)

An incorrect answer is provided.

43

A customer orders a large pizza with 2 toppings and a medium pizza with 2 toppings at a restaurant. The prices for the pizzas and toppings are listed below.

- Large pizza without toppings: $12.50
- Medium pizza without toppings: $10.25
- Toppings: $1.25 each

A 20% discount is applied to the customer's order. What is the total price of the order, excluding tax, after the discount?

Show your work.

Answer $ ______________

EXEMPLARY RESPONSE

43

A customer orders a large pizza with 2 toppings and a medium pizza with 2 toppings at a restaurant. The prices for the pizzas and toppings are listed below.

- Large pizza without toppings: \$12.50
- Medium pizza without toppings: \$10.25
- Toppings: \$1.25 each

A 20% discount is applied to the customer's order. What is the total price of the order, excluding tax, after the discount?

Show your work.

$12.50 + 2(1.25) = 15.00$

$10.25 + 2(1.25) = 12.75$

$(0.80)(15 + 12.75) = (0.80)(27.75) = 22.20$

Answer \$ 22.20

GUIDE PAPER 1

43

A customer orders a large pizza with 2 toppings and a medium pizza with 2 toppings at a restaurant. The prices for the pizzas and toppings are listed below.

- Large pizza without toppings: $12.50
- Medium pizza without toppings: $10.25
- Toppings: $1.25 each

A 20% discount is applied to the customer's order. What is the total price of the order, excluding tax, after the discount?

Show your work.

medium pizza = 10.25
large pizza = 12.50
toppings = 1.25 each
1.25 x 4 = 5
12.50 plus 10.25 = 22.75
22.75 + 5 = 27.75
27.75 x 20% = 5.55
27.75 - 5.55 = 22.20

Answer $ 22.20

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The total price of the two pizzas with toppings including the discount is correctly determined using a mathematically sound procedure.

This response is complete and correct.

GUIDE PAPER 2

43

A customer orders a large pizza with 2 toppings and a medium pizza with 2 toppings at a restaurant. The prices for the pizzas and toppings are listed below.

- Large pizza without toppings: $12.50
- Medium pizza without toppings: $10.25
- Toppings: $1.25 each

A 20% discount is applied to the customer's order. What is the total price of the order, excluding tax, after the discount?

Show your work.

1st.
1.25x2=2.5
12.5+2.5=15
15x0.2=3
15-3=12
$12
2nd.
1.25x2=2.5
10.25+2.5=12.75
12.75x0.2=2.55
12.75-2.55=10.20
$10.20
10.20+12=22.20

Answer $ 22.20

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The total price of the two pizzas with toppings including the discount is correctly determined using a mathematically sound procedure.

This response is complete and correct.

GUIDE PAPER 3

43

A customer orders a large pizza with 2 toppings and a medium pizza with 2 toppings at a restaurant. The prices for the pizzas and toppings are listed below.

- Large pizza without toppings: $12.50
- Medium pizza without toppings: $10.25
- Toppings: $1.25 each

A 20% discount is applied to the customer's order. What is the total price of the order, excluding tax, after the discount?

Show your work.

$$12.50 + 1.25 + 1.25 = 15.00$$

$$10.25 + 1.25 + 1.25 = 12.75$$

$$15 + 12.75 = 27.75$$

$$\frac{80}{100} = \frac{22.20}{27.75}$$

Answer $ 22.20

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The total price of the two pizzas with toppings including the discount is correctly determined using a mathematically sound procedure.

This response is complete and correct.

GUIDE PAPER 4

43

A customer orders a large pizza with 2 toppings and a medium pizza with 2 toppings at a restaurant. The prices for the pizzas and toppings are listed below.

- Large pizza without toppings: $12.50
- Medium pizza without toppings: $10.25
- Toppings: $1.25 each

A 20% discount is applied to the customer's order. What is the total price of the order, excluding tax, after the discount?

Show your work.

2 x 1.25= 2.50
12.50 + 2.50= $15

10.25 + 2.50 =$12.75

15+12.75= 27.75

Part = X
Whole= 27.75
Percent=100-20=80

X=(27.75)*(0.8)

X=$22.2

Tax

100+20=1.2

Part= X
Whole = 22.2
Percent = 1.2

X= (22.2)*(1.2)

X=26.64

Answer $ 26.64

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- An appropriate process is provided to determine the total price of the two pizzas with toppings including the discount.
- However, a twenty percent tax is inappropriately applied to the discounted price, resulting in an incorrect solution.

This response correctly addresses only some elements of the task.

GUIDE PAPER 5

43

A customer orders a large pizza with 2 toppings and a medium pizza with 2 toppings at a restaurant. The prices for the pizzas and toppings are listed below.

- Large pizza without toppings: $12.50
- Medium pizza without toppings: $10.25
- Toppings: $1.25 each

A 20% discount is applied to the customer's order. What is the total price of the order, excluding tax, after the discount?

Show your work.

12.50 + 10.25 + 2(1.25) =
$25.25

0.80 * 25.25 = $20.20

Answer $ 20.20

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- An error occurs when calculating the price of the two pizzas with toppings.
- However, the rest of the work to apply the twenty percent discount is carried out correctly.

This response correctly addresses only some elements of the task.

GUIDE PAPER 6

43

A customer orders a large pizza with 2 toppings and a medium pizza with 2 toppings at a restaurant. The prices for the pizzas and toppings are listed below.

- Large pizza without toppings: $12.50
- Medium pizza without toppings: $10.25
- Toppings: $1.25 each

A 20% discount is applied to the customer's order. What is the total price of the order, excluding tax, after the discount?

Show your work.

12.50+ 2.50 =15
10.25+2.50=12.75
=27.75

Answer $ 22.20

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- The price of the two pizzas with toppings is calculated correctly.
- However, it is unclear how the solution is obtained.

This response contains the correct solution, but the required work is incomplete.

GUIDE PAPER 7

43

A customer orders a large pizza with 2 toppings and a medium pizza with 2 toppings at a restaurant. The prices for the pizzas and toppings are listed below.

- Large pizza without toppings: $12.50
- Medium pizza without toppings: $10.25
- Toppings: $1.25 each

A 20% discount is applied to the customer's order. What is the total price of the order, excluding tax, after the discount?

Show your work.

12.50 + 10.25=22.75 22.75 ×
1.25=28.4375

Answer $ 28.4375

Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- An incorrect procedure is used to determine an incorrect solution.

Holistically, this response shows no overall understanding of the task.

43

A customer orders a large pizza with 2 toppings and a medium pizza with 2 toppings at a restaurant. The prices for the pizzas and toppings are listed below.

- Large pizza without toppings: \$12.50
- Medium pizza without toppings: \$10.25
- Toppings: \$1.25 each

A 20% discount is applied to the customer's order. What is the total price of the order, excluding tax, after the discount?

Show your work.

12.50 + 10.25 +1.25= 24.00

Answer \$ 24.00

Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- The twenty percent discount is not addressed, and the price of the two pizzas and one topping is inappropriately provided as an incorrect solution.

Holistically, this response shows no overall understanding of the task.

44

Jim and Devin ride their bicycles to meet at a friend's house.

- Jim rides $3\frac{1}{3}$ miles in $\frac{1}{2}$ hour.
- Devin rides $1\frac{7}{8}$ miles in $\frac{1}{4}$ hour.

What is the difference, in miles per hour, between the average speeds of Jim and Devin?

Explain how you determined your answer.

EXEMPLARY RESPONSE

44

Jim and Devin ride their bicycles to meet at a friend's house.

- Jim rides $3\frac{1}{3}$ miles in $\frac{1}{2}$ hour.
- Devin rides $1\frac{7}{8}$ miles in $\frac{1}{4}$ hour.

What is the difference, in miles per hour, between the average speeds of Jim and Devin?

Explain how you determined your answer.

The difference between the two travel rates is $^5/_6$ mile per hour.
I know this is true because I can divide the number of miles
by the fraction of time for both riders
and get the number of miles per hour for each and subtract the rates.

When I do this,
I get $^{10}/_3 \div {}^1/_2 = {}^{10}/_3 \times {}^2/_1 = {}^{20}/_3$ for Jim
and $^{15}/_8 \div {}^1/_4 = {}^{15}/_8 \times {}^4/_1 = {}^{60}/_8 = {}^{15}/_2$ for Devin.

Then, I find a common denominator to subtract them
and I get $^{45}/_6 - {}^{40}/_6 = {}^5/_6$
which is a $^5/_6$ mile per hour difference.

OR other valid response

44

Jim and Devin ride their bicycles to meet at a friend's house.

- Jim rides $3\frac{1}{3}$ miles in $\frac{1}{2}$ hour.
- Devin rides $1\frac{7}{8}$ miles in $\frac{1}{4}$ hour.

What is the difference, in miles per hour, between the average speeds of Jim and Devin?

Explain how you determined your answer.

$$\frac{m}{h} = \frac{3\frac{1}{3}}{\frac{1}{2}} = \frac{x}{1}$$

$$6\frac{2}{3} = 15$$

$$\frac{m}{h} = \frac{1\frac{7}{8}}{\frac{1}{4}} = \frac{x}{1}$$

$$7\frac{1}{2} = 1D$$

$$\begin{array}{r} 7\frac{1}{2} \\ -6\frac{2}{3} \\ \hline \end{array} \qquad \frac{5}{6} -$$

The diffrence in speed is $\frac{5}{6}$

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The difference between the average speeds is correctly determined and explained using mathematically sound procedures.

The response is complete and correct.

GUIDE PAPER 2

44

Jim and Devin ride their bicycles to meet at a friend's house.

- Jim rides $3\frac{1}{3}$ miles in $\frac{1}{2}$ hour.
- Devin rides $1\frac{7}{8}$ miles in $\frac{1}{4}$ hour.

What is the difference, in miles per hour, between the average speeds of Jim and Devin?

Explain how you determined your answer.

$3\frac{1}{3} \times 2 = 6\frac{2}{3}$ this is the miles per hour for Jim

$1\frac{7}{8} \times 4 = 7\frac{1}{2}$ this is the miles per hour for Devin

$7\frac{1}{2} - 6\frac{2}{3} = \frac{5}{6}$ this is the difference between their miles per hour.

I first found the miles per hour for both of them ($6\frac{2}{3}$ for Jim and $7\frac{1}{2}$ for Devin) I then found the difference between the two miles per hour.

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The difference between the average speeds is correctly determined and explained using mathematically sound procedures.

The response is complete and correct.

GUIDE PAPER 3

44

Jim and Devin ride their bicycles to meet at a friend's house.

- Jim rides $3\frac{1}{3}$ miles in $\frac{1}{2}$ hour.
- Devin rides $1\frac{7}{8}$ miles in $\frac{1}{4}$ hour.

What is the difference, in miles per hour, between the average speeds of Jim and Devin?

Explain how you determined your answer.

$\frac{5}{6}$ is the difference between the average speeds of Jim and Devin. Devin: $3\frac{1}{3} \div \frac{1}{2} = 6\frac{2}{3}$ mph. Jim: $1\frac{7}{8} \div \frac{1}{4} = 7\frac{1}{2}$ mph. $7\frac{1}{2}$ mph - $6\frac{2}{3} = \frac{5}{6}$.

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The difference between the average speeds is correctly determined and explained using mathematically sound procedures.
- Although the labeling for Devin and Jim's work is reversed, it does not detract from demonstrating a thorough understanding.

This response contains sufficient work to demonstrate a thorough understanding.

GUIDE PAPER 4

44

Jim and Devin ride their bicycles to meet at a friend's house.

- Jim rides $3\frac{1}{3}$ miles in $\frac{1}{2}$ hour.
- Devin rides $1\frac{7}{8}$ miles in $\frac{1}{4}$ hour.

What is the difference, in miles per hour, between the average speeds of Jim and Devin?

Explain how you determined your answer.

$$1\frac{7}{8} \times 2 = 3\frac{6}{8} - 3\frac{1}{3} = \frac{5}{12}$$

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- The difference in the distance traveled per half-hour is determined.
- However, it is inappropriately provided as an incorrect solution.

The response correctly addresses only some elements of the task.

GUIDE PAPER 5

44

Jim and Devin ride their bicycles to meet at a friend's house.

- Jim rides $3\frac{1}{3}$ miles in $\frac{1}{2}$ hour.
- Devin rides $1\frac{7}{8}$ miles in $\frac{1}{4}$ hour.

What is the difference, in miles per hour, between the average speeds of Jim and Devin?

Explain how you determined your answer.

jim: $3\frac{1}{3} \div \frac{1}{2} = \frac{20}{3}$

devin: $1\frac{7}{8} \div \frac{1}{4} = \frac{15}{2}$

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- The average speeds per hour for Jim and Devin are correctly determined.
- However, the difference between the speeds is not addressed.

The response correctly addresses only some elements of the task.

GUIDE PAPER 6

44

Jim and Devin ride their bicycles to meet at a friend's house.

- Jim rides $3\frac{1}{3}$ miles in $\frac{1}{2}$ hour.
- Devin rides $1\frac{7}{8}$ miles in $\frac{1}{4}$ hour.

What is the difference, in miles per hour, between the average speeds of Jim and Devin?

Explain how you determined your answer.

5/6 miles per hour
I determined my answer by finding their average speed in miles per hour, then subtracting the two to find the difference.

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- Although the correct difference between the average speeds is provided, the explanation is incomplete.

This response contains the correct solution but the required work is incomplete.

GUIDE PAPER 7

44

Jim and Devin ride their bicycles to meet at a friend's house.

- Jim rides $3\frac{1}{3}$ miles in $\frac{1}{2}$ hour.
- Devin rides $1\frac{7}{8}$ miles in $\frac{1}{4}$ hour.

What is the difference, in miles per hour, between the average speeds of Jim and Devin?

Explain how you determined your answer.

5/6

Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- The correct solution is provided with no explanation.

Per Scoring Policy #3 for 2- and 3-credit responses, this response receives no credit.

44

Jim and Devin ride their bicycles to meet at a friend's house.

- Jim rides $3\frac{1}{3}$ miles in $\frac{1}{2}$ hour.
- Devin rides $1\frac{7}{8}$ miles in $\frac{1}{4}$ hour.

What is the difference, in miles per hour, between the average speeds of Jim and Devin?

Explain how you determined your answer.

$3\frac{1}{3} \times \frac{1}{2} =$

$1\frac{2}{3}$

$\frac{7}{8} \times \frac{1}{4} =$

$\frac{7}{32}$

the difference in miles per hour is jim rides more

Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- An incorrect procedure is used, and an incorrect solution is provided.

The response is insufficient to show any understanding.

45

A group of sixth-grade students recorded their favorite flavors of ice cream. The results are shown in the table below.

FAVORITE FLAVORS OF ICE CREAM

Flavor of Ice Cream	Number of Students
Vanilla	12
Chocolate	8
Strawberry	5
Pistachio	3
Other	4

Based on the results, what would be the predicted number of students in the entire sixth grade whose favorite ice cream flavor is chocolate, if there are a total of 120 students in the sixth grade?

Show your work.

Answer ________________ students

EXEMPLARY RESPONSE

45

A group of sixth-grade students recorded their favorite flavors of ice cream. The results are shown in the table below.

FAVORITE FLAVORS OF ICE CREAM

Flavor of Ice Cream	Number of Students
Vanilla	12
Chocolate	8
Strawberry	5
Pistachio	3
Other	4

Based on the results, what would be the predicted number of students in the entire sixth grade whose favorite ice cream flavor is chocolate, if there are a total of 120 students in the sixth grade?

Show your work.

$8 \div 32 = 0.25$

$0.25 \times 120 = 30$

OR other valid process

Answer 30 students

45

A group of sixth-grade students recorded their favorite flavors of ice cream. The results are shown in the table below.

FAVORITE FLAVORS OF ICE CREAM

Flavor of Ice Cream	Number of Students
Vanilla	12
Chocolate	8
Strawberry	5
Pistachio	3
Other	4

Based on the results, what would be the predicted number of students in the entire sixth grade whose favorite ice cream flavor is chocolate, if there are a total of 120 students in the sixth grade?

Show your work.

$\frac{8}{32} = \frac{x}{120}$

$32 \times 3.75 = 120$

8 x 3.75 = 30 students

Answer 30 students

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The predicted number of students preferring chocolate in the entire sixth grade is correctly determined using sound procedures.

This response is complete and correct.

GUIDE PAPER 2

45

A group of sixth-grade students recorded their favorite flavors of ice cream. The results are shown in the table below.

FAVORITE FLAVORS OF ICE CREAM

Flavor of Ice Cream	Number of Students
Vanilla	12
Chocolate	8
Strawberry	5
Pistachio	3
Other	4

Based on the results, what would be the predicted number of students in the entire sixth grade whose favorite ice cream flavor is chocolate, if there are a total of 120 students in the sixth grade?

Show your work.

12 + 8 + 5 + 3 + 4 = 32

Fraction of people whose favorite ice cream is chocolate: 8/32 ->simplified to 1/4.

1/4 of 120 students is 30.

Answer 30 students

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The predicted number of students preferring chocolate in the entire sixth grade is correctly determined using sound procedures.

This response is complete and correct.

GUIDE PAPER 3

45

A group of sixth-grade students recorded their favorite flavors of ice cream. The results are shown in the table below.

FAVORITE FLAVORS OF ICE CREAM

Flavor of Ice Cream	Number of Students
Vanilla	12
Chocolate	8
Strawberry	5
Pistachio	3
Other	4

Based on the results, what would be the predicted number of students in the entire sixth grade whose favorite ice cream flavor is chocolate, if there are a total of 120 students in the sixth grade?

Show your work.

120 ÷ 32=3.75
3.75 × 8=30
Thirty kids

Answer 30 students

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The predicted number of students preferring chocolate in the entire sixth grade is correctly determined using sound procedures.

This response is complete and correct.

GUIDE PAPER 4

45

A group of sixth-grade students recorded their favorite flavors of ice cream. The results are shown in the table below.

FAVORITE FLAVORS OF ICE CREAM

Flavor of Ice Cream	Number of Students
Vanilla	12
Chocolate	8
Strawberry	5
Pistachio	3
Other	4

Based on the results, what would be the predicted number of students in the entire sixth grade whose favorite ice cream flavor is chocolate, if there are a total of 120 students in the sixth grade?

Show your work.

120 ÷ 32 = 3.75 × 4=15
it would be 15 kids favorite ice cream

Answer 15 students

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- The value for the ratio of the entire class size to the survey group size is correctly determined.
- However, this value is not applied correctly, resulting in an incorrect solution.

This response correctly addresses only some elements of the task.

GUIDE PAPER 5

45

A group of sixth-grade students recorded their favorite flavors of ice cream. The results are shown in the table below.

FAVORITE FLAVORS OF ICE CREAM

Flavor of Ice Cream	Number of Students
Vanilla	12
Chocolate	8
Strawberry	5
Pistachio	3
Other	4

Based on the results, what would be the predicted number of students in the entire sixth grade whose favorite ice cream flavor is chocolate, if there are a total of 120 students in the sixth grade?

Show your work.

120 ÷ 32=4
4x8=32

Answer 32 students

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- An appropriate process is provided to determine the predicted number of students preferring chocolate in the entire sixth grade.
- However, a calculation error occurs, resulting in an incorrect solution.

This response contains an incorrect solution but applies a mathematically appropriate process.

GUIDE PAPER 6

45

A group of sixth-grade students recorded their favorite flavors of ice cream. The results are shown in the table below.

FAVORITE FLAVORS OF ICE CREAM

Flavor of Ice Cream	Number of Students
Vanilla	12
Chocolate	8
Strawberry	5
Pistachio	3
Other	4

Based on the results, what would be the predicted number of students in the entire sixth grade whose favorite ice cream flavor is chocolate, if there are a total of 120 students in the sixth grade?

Show your work.

12+8+5+3+4=32
120/32=3.75

Answer 3.75 students

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- The value for the ratio of the entire class size to the survey group size is correctly determined.
- However, this value is not further applied and is inappropriately provided as the solution.

This response correctly addresses only some elements of the task.

GUIDE PAPER 7

45

A group of sixth-grade students recorded their favorite flavors of ice cream. The results are shown in the table below.

FAVORITE FLAVORS OF ICE CREAM

Flavor of Ice Cream	Number of Students
Vanilla	12
Chocolate	8
Strawberry	5
Pistachio	3
Other	4

Based on the results, what would be the predicted number of students in the entire sixth grade whose favorite ice cream flavor is chocolate, if there are a total of 120 students in the sixth grade?

Show your work.

12+8+5+3+4= 32 32 ÷ 5 = 6.4 6.6 × 120 = 792

Answer 792 students

Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- An incorrect procedure is used and an incorrect solution is provided.

Holistically, this response shows no overall understanding of the task.

GUIDE PAPER 8

45

A group of sixth-grade students recorded their favorite flavors of ice cream. The results are shown in the table below.

FAVORITE FLAVORS OF ICE CREAM

Flavor of Ice Cream	Number of Students
Vanilla	12
Chocolate	8
Strawberry	5
Pistachio	3
Other	4

Based on the results, what would be the predicted number of students in the entire sixth grade whose favorite ice cream flavor is chocolate, if there are a total of 120 students in the sixth grade?

Show your work.

12+8+5+3+4=32-120=88

Answer 88 students

Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

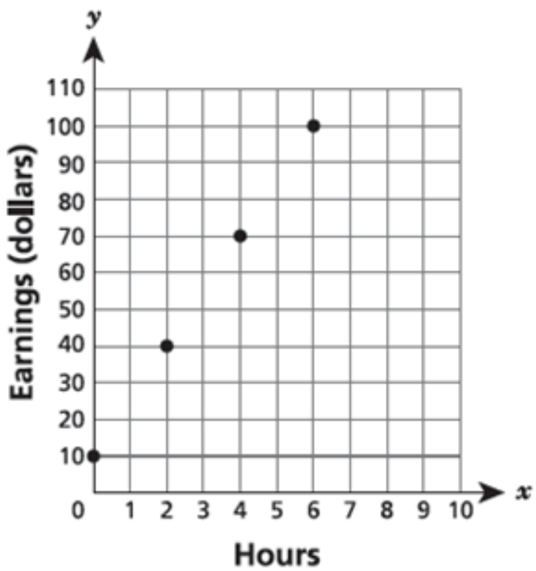
- An incorrect procedure is used and an incorrect solution is provided.

Holistically, this response shows no overall understanding of the task.

46

Eric's and Jenna's earnings at their jobs are shown in the graph and the table below.

ERIC'S EARNINGS



JENNA'S EARNINGS

Hours, x	Earnings (dollars), y
3	54
6	108
9	162
12	216

For which person is the amount of earnings proportional to the number of hours worked?

Explain your answer.

____

____

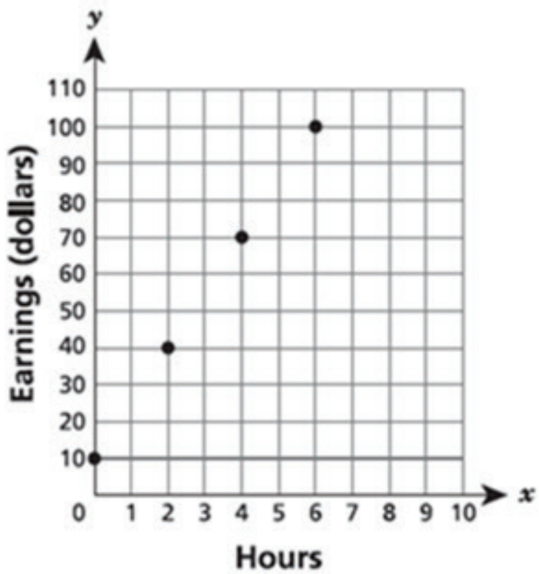
____

EXEMPLARY RESPONSE

46

Eric's and Jenna's earnings at their jobs are shown in the graph and the table below.

ERIC'S EARNINGS



JENNA'S EARNINGS

Hours, x	Earnings (dollars), y
3	54
6	108
9	162
12	216

For which person is the amount of earnings proportional to the number of hours worked?

Explain your answer.

Jenna's earnings show a proportional relationship because the amount of earnings for each row is always 18 times the corresponding number of hours worked.

$^{54}/_{3} = {}^{108}/_{6} = {}^{162}/_{9} = {}^{216}/_{12} = 18$

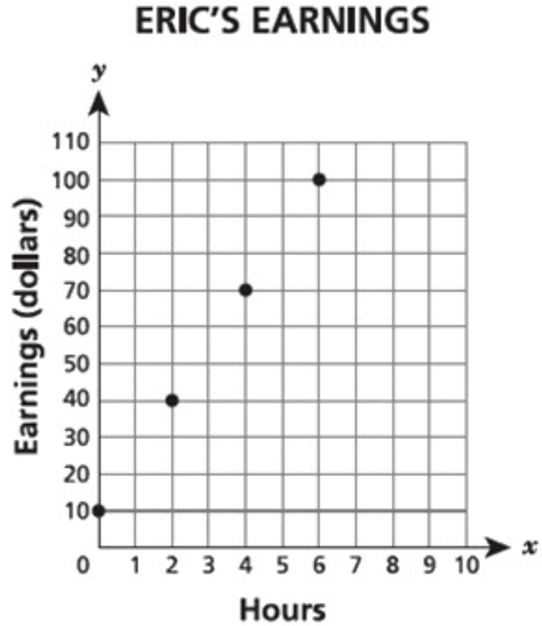
Eric's earnings are not proportional.

$^{40}/_{2} = 20$
$^{70}/_{4} = 17.5$
$^{100}/_{6} = 16\ ^{2}/_{3}$

OR other valid response

46

Eric's and Jenna's earnings at their jobs are shown in the graph and the table below.



JENNA'S EARNINGS

Hours, x	Earnings (dollars), y
3	54
6	108
9	162
12	216

For which person is the amount of earnings proportional to the number of hours worked?

Explain your answer.

Jenna's earnings are proportional to how many hours she worked. This is because the quotient of how many dollars she makes divided by how many hours she worked is always the same. For example, 54/3 is 18, 108/6 is 18, 162/9 is 18, and 216/12 is 18. Therefore, her earnings are proportional, and the rate of change is $18.00 per hour. Eric's earnings are not proportional because after 0 hours of working, he has not made $0.00. Instead, he has made $10.00. Therefore, Jenna's earnings are proportional, and Eric's are not.

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The correct person is identified, and a valid explanation is provided.

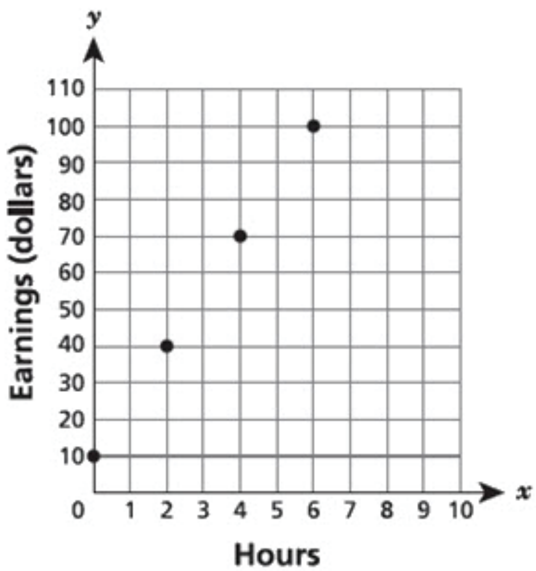
This response is complete and correct.

GUIDE PAPER 2

46

Eric's and Jenna's earnings at their jobs are shown in the graph and the table below.

ERIC'S EARNINGS



JENNA'S EARNINGS

Hours, x	Earnings (dollars), y
3	54
6	108
9	162
12	216

For which person is the amount of earnings proportional to the number of hours worked?

Explain your answer.

Jenna's earnings are proportional. The unit of proportionality is \$18 an hour (52/3=18). Multiplying any unit of hours on her chart with \$18 you will get her written (y) vaulue each time.

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The correct person is identified, and a valid explanation is provided.
- Although an incorrect equation is written (52/3=18), the remaining explanation demonstrates understanding that the constant of proportionality is the same for all rows in the table.

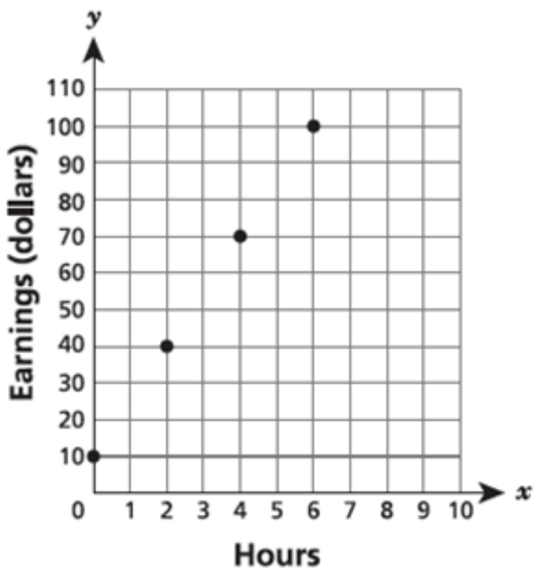
This response is sufficient to demonstrate a thorough understanding.

GUIDE PAPER 3

46

Eric's and Jenna's earnings at their jobs are shown in the graph and the table below.

ERIC'S EARNINGS



JENNA'S EARNINGS

Hours, x	Earnings (dollars), y
3	54
6	108
9	162
12	216

For which person is the amount of earnings proportional to the number of hours worked?

Explain your answer.

Jenna's Earnings are proportional to the number of hours worked but Eric's earnings are not. On a graph it is proportional if there is a straight line passing through the origin, even though for Eric's earnings it is a straight line it didnt pass throught the origin so it cannot be proportional. However Jenna's Earnings are proportional because there is a unit rate of 18 dollars per hour worked.

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The correct person is identified, and a valid explanation is provided.

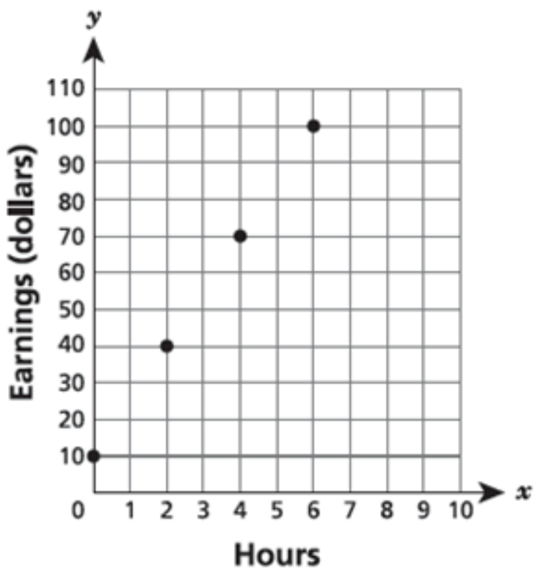
This response is sufficient to demonstrate a thorough understanding.

GUIDE PAPER 4

46

Eric's and Jenna's earnings at their jobs are shown in the graph and the table below.

ERIC'S EARNINGS



JENNA'S EARNINGS

Hours, x	Earnings (dollars), y
3	54
6	108
9	162
12	216

For which person is the amount of earnings proportional to the number of hours worked?

Explain your answer.

Jenna's earnings are proportional because she earns 18 dollars per hour. Erics earnings are not proportional because he earns a different amount of money per x hours.

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- The correct person is identified.
- However, the explanation provided is incomplete; it is unclear how the unit rate was obtained.

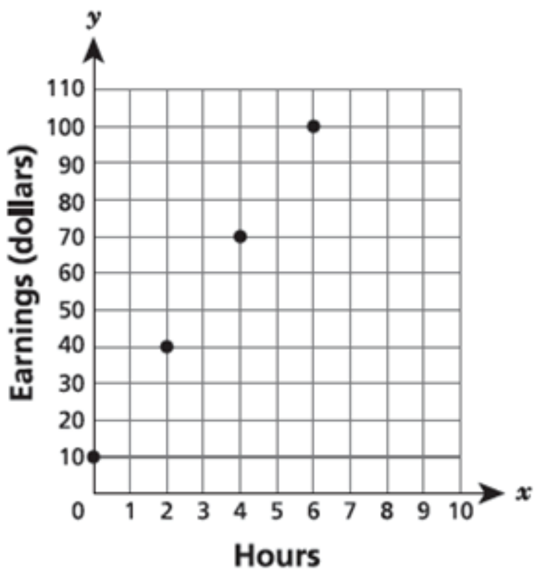
This response correctly addresses only some elements of the task.

GUIDE PAPER 5

46

Eric's and Jenna's earnings at their jobs are shown in the graph and the table below.

ERIC'S EARNINGS



JENNA'S EARNINGS

Hours, x	Earnings (dollars), y
3	54
6	108
9	162
12	216

For which person is the amount of earnings proportional to the number of hours worked?

Explain your answer.

if you divide the earnings by the hours of jennas work for each ratio you can see that it is proportional, but with eric it starts with him earning 10 for 0 hours and then jumping to 40 for 2 hours, do then keep going without any pattern. jenny's earnings are proportional to the hours worked

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- The correct person is identified.
- However, the explanation provided is incomplete.

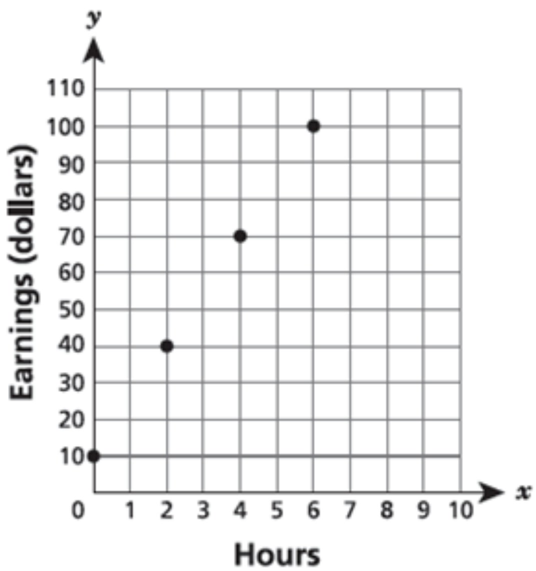
This response correctly addresses only some elements of the task.

GUIDE PAPER 6

46

Eric's and Jenna's earnings at their jobs are shown in the graph and the table below.

ERIC'S EARNINGS



JENNA'S EARNINGS

Hours, x	Earnings (dollars), y
3	54
6	108
9	162
12	216

For which person is the amount of earnings proportional to the number of hours worked?

Explain your answer.

18 dollors is the amount proportional to the number of hour worked

$54 \div 3 = 18$

$108 \div 6 = 18$

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- Although the constant of proportionality is correctly calculated for 3 and 6 hours, the calculations for 9 and 12 hours are not addressed.
- No correct identification is made.

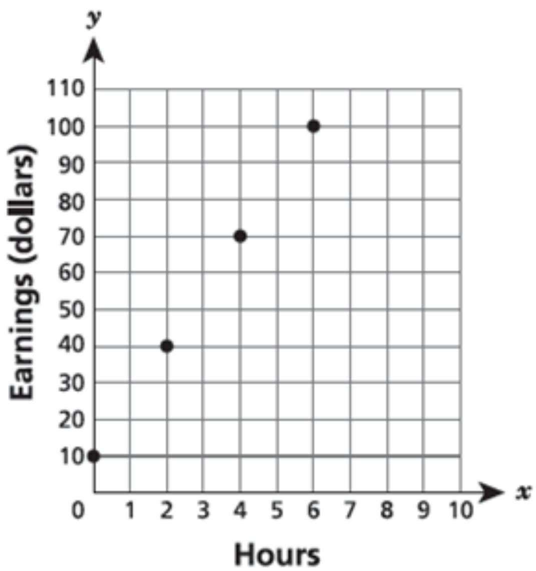
This response correctly addresses only some elements of the task.

GUIDE PAPER 7

46

Eric's and Jenna's earnings at their jobs are shown in the graph and the table below.

ERIC'S EARNINGS



JENNA'S EARNINGS

Hours, x	Earnings (dollars), y
3	54
6	108
9	162
12	216

For which person is the amount of earnings proportional to the number of hours worked?

Explain your answer.

Jenna becuase 100 ÷ 6=16.67 and 108 ÷ 6=18.

Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- Although some elements are correct, the explanation is insufficient to show proportionality.

Holistically, the response is insufficient to show any understanding.

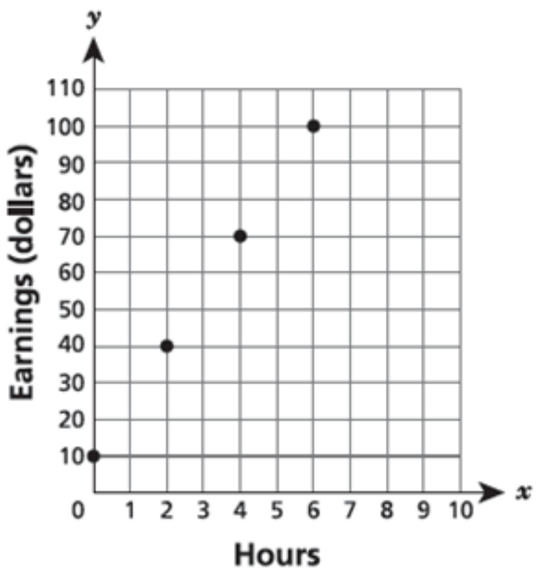
GUIDE PAPER 8

Additional

46

Eric's and Jenna's earnings at their jobs are shown in the graph and the table below.

ERIC'S EARNINGS



JENNA'S EARNINGS

Hours, x	Earnings (dollars), y
3	54
6	108
9	162
12	216

For which person is the amount of earnings proportional to the number of hours worked?

Explain your answer.

the table is proportional.

Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- An identification of "the table" is made; however, no explanation is provided.

Per Scoring Policy #3 for 2- and 3-credit responses, this response receives no credit.

47

An expression is shown below.

$$\left(\frac{1}{2}\right)(-0.4) \div \left(\frac{1}{3}\right)$$

Evaluate the expression.

Show your work.

Answer ________________

EXEMPLARY RESPONSE

47

An expression is shown below.

$$\left(\frac{1}{2}\right)(-0.4) \div \left(\frac{1}{3}\right)$$

Evaluate the expression.

Show your work.

$^{1}/_{2}\ (^{-4}/_{10}) \div {}^{1}/_{3}$

$^{-1}/_{5} \div {}^{1}/_{3}$

$^{-1}/_{5} \times {}^{3}/_{1}$

$^{-3}/_{5}$

Answer $^{-3}/_{5}$ or $^{-6}/_{10}$

OR other valid response

47

An expression is shown below.

$$\left(\frac{1}{2}\right)(-0.4) \div \left(\frac{1}{3}\right)$$

Evaluate the expression.

Show your work.

The first thing I did was convert (-0.4) into a fraction so that I could work with all fractions.

-0.4 as a fraction is $-\frac{2}{5}$.

Then I solved:

$\left(\frac{1}{2}\right) \times \left(-\frac{2}{5}\right)$ is $-\frac{2}{10}$ which can be simplified into $-\frac{1}{5}$.

Lastly, I did $-\frac{1}{5} \div \frac{1}{3}$ which is $-\frac{3}{5}$

Answer $-\frac{3}{5}$

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The value of the expression is correctly determined using a mathematically sound procedure.

This response is complete and correct.

47

An expression is shown below.

$$\left(\frac{1}{2}\right)(-0.4) \div \left(\frac{1}{3}\right)$$

Evaluate the expression.

Show your work.

$\left(\frac{1}{2}\right)$ x -0.4=-0.2

-0.2 ÷ $\frac{1}{3}$ =-0.6

Answer -0.6

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The value of the expression is correctly determined using a mathematically sound procedure.

This response is complete and correct.

GUIDE PAPER 3

47

An expression is shown below.

$$\left(\frac{1}{2}\right)(-0.4) \div \left(\frac{1}{3}\right)$$

Evaluate the expression.

Show your work.

.5 times -0.4 = -0.2 divided by $\frac{1}{3}$ = -0.6

Answer -0.6

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The value of the expression is correctly determined using a mathematically sound procedure.
- The run-on equation does not detract from the demonstration of a thorough understanding.

This response contains sufficient work to demonstrate a thorough understanding.

GUIDE PAPER 4

47

An expression is shown below.

$$\left(\frac{1}{2}\right)(-0.4) \div \left(\frac{1}{3}\right)$$

Evaluate the expression.

Show your work.

0.5x-0.4 ÷ .33

$$\begin{array}{r} .5 \\ \times \quad -.4 \\ \hline -.2 \end{array} \div .33$$

-.2 ÷ .33=-0.60606060606

Answer -0.60606060606

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- A mathematically sound procedure is used to evaluate the expression.
- However, the conversion of the fraction 1/3 is incorrectly truncated to .33, resulting in an incorrect solution.

This response contains an incorrect solution but applies a mathematically appropriate process.

GUIDE PAPER 5

47

An expression is shown below.

$$\left(\frac{1}{2}\right)(-0.4) \div \left(\frac{1}{3}\right)$$

Evaluate the expression.

Show your work.

$(\frac{1}{2})(-0.4)=-0.2$

$-0.2 \div (\frac{1}{3})=-1.2$

Answer -1.2

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- A mathematically sound procedure is used to evaluate the expression.
- However, a calculation error occurs in the division step, resulting in an incorrect solution.

This response contains an incorrect solution but applies a mathematically appropriate process.

GUIDE PAPER 6

47

An expression is shown below.

$$\left(\frac{1}{2}\right)(-0.4) \div \left(\frac{1}{3}\right)$$

Evaluate the expression.

Show your work.

$-0.4 \div \dfrac{1}{3} = -1.2$ $\qquad$ $\dfrac{1}{2} - 1.2 = -0.7$

Answer -0.7

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- The division step is calculated correctly.
- However, an operation error occurs in the second step (subtraction instead of multiplication), resulting in an incorrect solution.

This response correctly addresses only some elements of the task.

47

An expression is shown below.

$$\left(\frac{1}{2}\right)(-0.4) \div \left(\frac{1}{3}\right)$$

Evaluate the expression.

Show your work.

$\frac{1}{2}$ x -0.4 / $\frac{1}{3}$

Answer $\frac{-1}{15}$

Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- The value of the expression is incorrectly determined, and no work is shown.

This response is incorrect, and, holistically, is insufficient to show any understanding.

47

An expression is shown below.

$$\left(\frac{1}{2}\right)(-0.4) \div \left(\frac{1}{3}\right)$$

Evaluate the expression.

Show your work.

$$\frac{1}{2} \text{ x } -.4 \div \frac{1}{3} = -.6 = -\frac{3}{5}$$

Answer $-.6 = -\frac{3}{5}$

Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- The correct solution is provided with no work.

Per Scoring Policy #3 for 2- and 3-credit responses, this response receives no credit.

48

Nicholas went to a bowling alley. The bowling alley charged Nicholas a one-time fee of \$3.75 to rent shoes and \$5.25 for each game he played. Write and solve an equation to determine the number of games, x, that Nicholas played if he spent a total of \$30.00 on shoes and games.

Show your work.

Answer ________________ games

EXEMPLARY RESPONSE

48

Nicholas went to a bowling alley. The bowling alley charged Nicholas a one-time fee of \$3.75 to rent shoes and \$5.25 for each game he played. Write and solve an equation to determine the number of games, x, that Nicholas played if he spent a total of \$30.00 on shoes and games.

Show your work.

Let x equal the number of games.

$3.75 + 5.25x = 30$

$5.25x = 30 - 3.75$

$5.25x = 26.25$

$x = 26.25 \div 5.25$

$x = 5$

5 games

OR other valid process

Answer 5 games

48

Nicholas went to a bowling alley. The bowling alley charged Nicholas a one-time fee of \$3.75 to rent shoes and \$5.25 for each game he played. Write and solve an equation to determine the number of games, x, that Nicholas played if he spent a total of \$30.00 on shoes and games.

Show your work.

Let 3.75= fee to rent shoes
Let 5.25= cost of each game
Let \$30= the total
Let x= the number of games he played
3.75+5.25x=30

$$3.75 + 5.25x = 30$$
$$-3.75 \quad -3.75$$
$$\frac{5.25x}{5.25} = \frac{26.25}{5.25}$$
$$x = 5$$

Answer 5 games

Score Credit 3 (out of 3 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- A correct equation is written, and it is correctly solved to determine the number of games.

This response is complete and correct.

GUIDE PAPER 2

48

Nicholas went to a bowling alley. The bowling alley charged Nicholas a one-time fee of \$3.75 to rent shoes and \$5.25 for each game he played. Write and solve an equation to determine the number of games, x, that Nicholas played if he spent a total of \$30.00 on shoes and games.

Show your work.

$3.75+5.25x=30$

$$\begin{array}{r} 30 \\ -\ 3.75 \\ \hline 26.25 \end{array} \quad \frac{26.25}{5.25} = 5$$

Answer 5 games

Score Credit 3 (out of 3 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- A correct equation is written, and it is correctly solved to determine the number of games.

This response is complete and correct.

GUIDE PAPER 3

48

Nicholas went to a bowling alley. The bowling alley charged Nicholas a one-time fee of \$3.75 to rent shoes and \$5.25 for each game he played. Write and solve an equation to determine the number of games, x, that Nicholas played if he spent a total of \$30.00 on shoes and games.

Show your work.

3.75+5.25x=30
-3.75 -3.75
5.25x=26.25 ÷ 5.25=5

Answer 5 games

Score Credit 3 (out of 3 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- A correct equation is written, and it is correctly solved to determine the number of games.
- The run-on equation does not detract from the demonstration of a thorough understanding.

This response contains sufficient work to demonstrate a thorough understanding.

GUIDE PAPER 4

48

Nicholas went to a bowling alley. The bowling alley charged Nicholas a one-time fee of \$3.75 to rent shoes and \$5.25 for each game he played. Write and solve an equation to determine the number of games, x, that Nicholas played if he spent a total of \$30.00 on shoes and games.

Show your work.

5.25x +3.75 = 30.00
30.00-3.75
26.75 ÷ 5.25
=5.10

Answer 5.10 games

Score Credit 2 (out of 3 credits)

This response demonstrates a partial understanding of the mathematical concepts and procedures in the task.

- A correct equation is written to determine the number of games.
- However, a calculation error occurs when subtracting the price of the shoes from the total spent, resulting in an incorrect solution.

This response contains an incorrect solution but applies sound procedures.

GUIDE PAPER 5

48

Nicholas went to a bowling alley. The bowling alley charged Nicholas a one-time fee of \$3.75 to rent shoes and \$5.25 for each game he played. Write and solve an equation to determine the number of games, x, that Nicholas played if he spent a total of \$30.00 on shoes and games.

Show your work.

X= Number of games played

3.75+5.25x=30

-3.75 -3.75

$\frac{3.75x}{3.75}$ $\frac{26.25}{3.75}$

=7

Answer Seven games games

Score Credit 2 (out of 3 credits)

This response demonstrates a partial understanding of the mathematical concepts and procedures in the task.

- A correct equation is written to determine the number of games.
- However, an error occurs (3.75x instead of 5.25x), resulting in an incorrect solution.

This response contains an incorrect solution but applies sound procedures.

GUIDE PAPER 6

48

Nicholas went to a bowling alley. The bowling alley charged Nicholas a one-time fee of \$3.75 to rent shoes and \$5.25 for each game he played. Write and solve an equation to determine the number of games, x, that Nicholas played if he spent a total of \$30.00 on shoes and games.

Show your work.

x5.25-3.75=30   x5.25=26.25 ÷ 5.25=5 games5

Answer 5 games

Score Credit 2 (out of 3 credits)

This response demonstrates a partial understanding of the mathematical concepts and procedures in the task.

- A mathematically sound procedure is used to determine the correct number of games.
- However, an incorrect equation is written (using – instead of +).

This response appropriately addresses most, but not all, aspects of the task.

GUIDE PAPER 7

48

Nicholas went to a bowling alley. The bowling alley charged Nicholas a one-time fee of \$3.75 to rent shoes and \$5.25 for each game he played. Write and solve an equation to determine the number of games, x, that Nicholas played if he spent a total of \$30.00 on shoes and games.

Show your work.

$5.25x + 3.75 = 30$

Answer ______ games

Score Credit 1 (out of 3 credits)

This response demonstrates only a limited understanding of the mathematical concepts and procedures in the task.

- A correct equation is written.
- However, no solution is determined.

This response addresses some elements of the task correctly, but the required work is incomplete.

48

Nicholas went to a bowling alley. The bowling alley charged Nicholas a one-time fee of \$3.75 to rent shoes and \$5.25 for each game he played. Write and solve an equation to determine the number of games, x, that Nicholas played if he spent a total of \$30.00 on shoes and games.

Show your work.

$$\frac{3.75}{3.75} + \frac{5.25x}{5.25} = \frac{30.00}{3.75}$$

$$x = 8$$

Answer X=8 games

Score Credit 1 (out of 3 credits)

This response demonstrates only a limited understanding of the mathematical concepts and procedures in the task.

- A correct equation is written to determine the number of games.
- However, an incorrect procedure is used, and an incorrect solution is provided.

This response exhibits multiple flaws related to misunderstanding important aspects of the task.

GUIDE PAPER 9

48

Nicholas went to a bowling alley. The bowling alley charged Nicholas a one-time fee of \$3.75 to rent shoes and \$5.25 for each game he played. Write and solve an equation to determine the number of games, x, that Nicholas played if he spent a total of \$30.00 on shoes and games.

Show your work.

30-3.75=27.25
27.25 ÷ 5.25=5

Answer 27.25=5x games

Score Credit 1 (out of 3 credits)

This response demonstrates a limited understanding of the mathematical concepts and procedures in the task.

- A correct process is used to determine the number of games.
- However, calculation errors occur, and an incorrect equation is inappropriately provided as the solution.

This response exhibits multiple flaws related to misunderstanding important aspects of the task.

GUIDE PAPER 10

48

Nicholas went to a bowling alley. The bowling alley charged Nicholas a one-time fee of $3.75 to rent shoes and $5.25 for each game he played. Write and solve an equation to determine the number of games, x, that Nicholas played if he spent a total of $30.00 on shoes and games.

Show your work.

$3.75 \times 5.25 = 19.68$

$30.00 - 19.68 = 10.32$

Answer 10.32 games

Score Credit 0 (out of 3 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts and procedures in the task.

- An incorrect procedure is used to determine an incorrect solution.

This response is incorrect, and, holistically, is insufficient to show any understanding.

48

Nicholas went to a bowling alley. The bowling alley charged Nicholas a one-time fee of \$3.75 to rent shoes and \$5.25 for each game he played. Write and solve an equation to determine the number of games, x, that Nicholas played if he spent a total of \$30.00 on shoes and games.

Show your work.

$3.75 \times 8 = 30.00$
$5.25 \times 5 = 26.25$

Answer 8 shoes and 5 games games

Score Credit 0 (out of 3 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.

- Although the correct number of games is provided as a solution, it is unclear how it is obtained, and an incorrect number of shoes is also provided as part of the solution.

Although some elements are correct, holistically they are not sufficient to demonstrate even a limited understanding of the mathematical concepts in the task.